Exercise 6.6 SOLUTION

Q.1. **Find G.M. between:**

i. **-2 and 8**

Sol: Here $a=-2$ and $b=8$

$$G=\pm\sqrt{ab}=\pm\sqrt{(-2)(8)}=\pm\sqrt{-16}=\pm 4i$$

ii. **$-2i$ and $8i$**

Sol: $$G=\pm\sqrt{ab}=\pm\sqrt{(-2i)(8i)}=\pm\sqrt{-16i^2}=\pm\sqrt{-16(-1)}=\pm\sqrt{16}=\pm 4$$

iii. **6 and 9**

Sol: $a=6$, $b=9$

$$G=\pm\sqrt{ab}=\pm\sqrt{(6)(9)}=\pm\sqrt{54}=\pm 3\sqrt{6}$$

Q.2. **Insert four real geometric means between 3 and 96 .**

Sol: Let G_1,G_2,G_3,G_4 are four G.M,s between 3& 96

$3,G_1,G_2,G_3,G_4,96$ are in G.P

$$a_1=3\ \&\ a^6=a_1r^5=96 \Rightarrow 3r^5=96 \Rightarrow r^5=\frac{96}{3}=32=2^5\Rightarrow r=2$$

Then $G_1=a_1r=3(2)=6$

$G_2=a_1r^2=3(2)^2=3(4)=12$

$G_3=a_1r^3=3(2)^3=3(8)=24$

$G_4=a_1r^4=3(2)^4=3(48)=48$ So, four G.M's are $b=6,12,24,48$

Q.3. **If both x and y are positive distinct real numbers, show that the geometric mean between x and y is less than their arithmetic mean.**

Sol: Given $x>0\&y>0$ then

Here $a=x,\&b=y$

$$G.M=\sqrt{xy}\ \&\ A.M=\frac{a+b}{2}=\frac{x+y}{2}$$

$$A.M-G.M=\frac{x+y}{2}-\sqrt{xy}=\frac{x+y-2\sqrt{xy}}{2}=\frac{\left(\sqrt{x}\right)^2+\left(\sqrt{y}\right)^2-2\sqrt{xy}}{2}=\frac{\left(\sqrt{x}-\sqrt{y}\right)^2}{2}=\frac{\left(\sqrt{x}-\sqrt{y}\right)^2}{2}>0$$

$$\Rightarrow A.M-G.M>0\Rightarrow A.M>G.M \Rightarrow \text{or}\ \ G.M<A.M$$

Q.4. **For what value of n, $\frac{a^n+b^n}{a^{n-1}+b^{n-1}}$ is the positive geometric mean between a and b?**

Sol: If $\frac{a^n+b^n}{a^{n-1}+b^{n-1}}$ be G.M between $a\&b$

Then $$\frac{a^n+b^n}{a^{n-1}+b^{n-1}}=\sqrt{ab}=(ab)^{1/2}\Rightarrow a^n+b^n=\left(a^{n-1}+b^{n-1}\right)a^{1/2}b^{1/2}$$

$$a^n+b^n=a^{n-1+1/2}b^{1/2}+a^{1/2}b^{n-1+1/2}\Rightarrow a^n+b^n=a^{n-1/2}b^{1/2}+a^{1/2}b^{n-1/2}$$

$$a^n-a^{n-1/2}b^{\frac{1}{2}}=a^{\frac{1}{2}}b^{n-1/2}-b^n\Rightarrow a^{n-1/2}\left(a^{1/2}-b^{1/2}\right)=b^{n-1/2}\left(a^{1/2}-b^{1/2}\right)\Rightarrow\frac{a^{n-1/2}}{b^{n-1/2}}=\frac{\left(a^{\frac{1}{2}}-b^{\frac{1}{2}}\right)}{\left(a^{\frac{1}{2}}-b^{\frac{1}{2}}\right)}$$

$$\Rightarrow\frac{a^{n-1/2}}{b^{n-1/2}}=1\Rightarrow\left(\frac{a}{b}\right)^{n-1/2}=1=\left(\frac{a}{b}\right)^0\Rightarrow n-\frac{1}{2}=0\Rightarrow n=\frac{1}{2}$$

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Q.5. **The A.M. of two positive integral numbers exceeds their (positive) G.M. by 2 and their sum is 20, find the numbers.**

Sol: **Condition I** $\Rightarrow \frac{a+b}{2} = \sqrt{ab} + 2$

'x' by 2 $\Rightarrow a+b = 2\sqrt{ab} + 4$ $\quad I$

Condition II $a+b = 20 \quad \Rightarrow a = 20-b$ $\quad II$

$20-b+b = 2\sqrt{(20-b)b} + 4$ (Put II in I)

$\Rightarrow 20-4 = 2\sqrt{20b-b^2} \Rightarrow 16 = 2\sqrt{20b-b^2} \Rightarrow 8 = \sqrt{20b-b^2}$

$64 = 20b-b^2 \Rightarrow b^2-20b+64 = 0$

$\Rightarrow b^2-16b-4b+64 = 0$

$b(b-16)-4(b-16) = 0$

$\Rightarrow (b-16)(b-4) = 0 \Rightarrow b-16 = 0$ or $b-4 = 0 \Rightarrow b = 16$ or $b = 4$

When $b = 16$ then $a = 20-16 = 4$

When $b = 4$ then $a = 20-4 = 16$

Hence two number are $4,16$ or $16,4$

Q.6. **The A.M. between two numbers is 5 and their (positive) G.M. is 4. Find the numbers.**

Sol: Suppose two number are a & b then

Condition I $\Rightarrow \frac{a+b}{2} = 5$

$\Rightarrow a+b = 10 \quad I \Rightarrow a = 10-b \quad II$

Condition II $\sqrt{ab} = 4 \quad \Rightarrow ab = 16 \quad III$

Put II in III $\Rightarrow (10-b)b = 16 \Rightarrow 10b-b^2 = 16$

$\Rightarrow b^2-10b+16 = 0 \Rightarrow b^2-8b-2b+16 = 0 \Rightarrow b(b-8)-2(b-8) = 0$

$(b-8)(b-2) = 0 \Rightarrow b-8 = 0$ or $b-2 = 0 \Rightarrow b = 8$ or $b = 2$

When $b = 8$ then $a = 10-8 = 2$

When $b = 2$ then $a = 10-2 = 8$

Hence two number are $2,8$ or $8,2$

Q.7. **The arithmetic mean between two positive numbers a and b is double their geometric mean. Prove that $a:b = 2+\sqrt{3}:2-\sqrt{3}$.**

Sol: $AM = \frac{a+b}{2}$ and $GM = \sqrt{ab}$

Given that $AM = 2 \times GM$

$\frac{a+b}{2} = 2\sqrt{ab} \Rightarrow \frac{(a+b)^2}{4} = 4ab$ (square both sides)

$(a+b)^2 = 16ab$

$a^2+2ab+b^2 = 16ab$

$a^2-14ab+b^2 = 0$

$\left(\frac{a}{b}\right)^2 - 14\left(\frac{a}{b}\right)+1 = 0$

Let $x = \frac{a}{b}$

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$x^2 - 14x + 1 = 0$

$a = 1,\ b = -14,\ c = 1$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-14) \pm \sqrt{(-14)^2 - 4(1)(1)}}{2(1)} = \frac{14 \pm \sqrt{196 - 4}}{2} = \frac{14 \pm \sqrt{192}}{2} = \frac{14 \pm 8\sqrt{3}}{2} = 7 \pm 4\sqrt{3}$$

$$\Rightarrow \frac{a}{b} = 7 + 4\sqrt{3} \text{ or } \frac{a}{b} = 7 - 4\sqrt{3}$$

Now lets simplify the target ratio

$2 + \sqrt{3} : 2 - \sqrt{3}$

$$\frac{2+\sqrt{3}}{2-\sqrt{3}} = \frac{(2+\sqrt{3})(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} = \frac{4+2\sqrt{3}+2\sqrt{3}+3}{4-3} = 7 + 4\sqrt{3} = \frac{a}{b}$$

Hence proved that $a : b = 2 + \sqrt{3} : 2 - \sqrt{3}$

Q.8 **If one geometric mean G and two arithmetic means p and q are inserted between two positive numbers, show that $G^2 = (2p - q)(2q - p)$.**

Sol: Let G be a GM between a and b then a, G, b forms a GP.

Therefore, common ratio is

$$\frac{G}{a} = \frac{b}{G} \Rightarrow G^2 = ab \text{------}(1)$$

Also, given that p and q are the two AM's between a and b.

Then

a, p, q, b are in AP.

Let's denote the common difference as d, then

$p = a + d$

$q = a + 2d$

$b = a + 3d$

From these equations, we can express d in terms a, p, q, and b:

$d = p - a$

$$d = \frac{q-a}{2} \Rightarrow d = \frac{b-a}{3}$$

Now, let's express b and a in terms of p and q.

From $p = a + d$ and $q = a + 2d$, we can eliminate a:

$q - p = (a + 2d) - (a + d) = d$

So, the common difference is $d = q - p$.

Now we can express a and b in terms of p and q:

$a = p - d = p - (q - p) = 2p - q$

$b = a + 3d = (2p - q) + 3(q - p) = 2p - q + 3q - 3p = 2q - p$

Now, recall from equation (1) that $G^2 = ab$.

Substituting the expressions for a and b in terms of p and q:

$G^2 = (2p - q)(2q - p)$

This is exactly what we needed to show.

Final Answer: The final answer is $G^2 = (2p - q)(2q - p)$

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